

RAMCO Institute of Technology, Rajapalayam
Department of Mechanical Engineering
Academic Year 2020-2021 (Odd Semester)

Innovative Practices(s) Description

Name of the Faculty : R.Prabhakaran, AP/Mechanical
Subject Code and Title : ME8593 – Design of Machine Elements
Year and Branch : III – Mechanical Engineering
Topics Covered : Tensile Test on Mild Steel
Name of the Laboratory : Strength of Material Laboratory
Website Link : <http://sm-nitk.vlabs.ac.in/exp13/index.html>
No.of Students participated : 32
Date & Time : 24.08.2020 & 10.40AM to 11.40AM

Demonstration of Tensile Test on Mild Steel

OBJECTIVE: To study the mechanical properties of Mild Steel specimen under tension load.

In tension test ends of a test piece are fixed into grips connected to a straining device and to a load measuring device. The test involves straining a test piece by tensile force generally to fracture for the purpose of determining one or more of the mechanical properties.

The students can able to evaluate Yield Stress, Tensile Strength, Modulus of Elasticity, Percentage Elongation and Percentage reduction in area after completion of this virtual laboratory.

OUTCOME : The Virtual Laboratory Experiments helps in attaining the following CO-PO mapping:

Course Outcome / Programme Outcomes	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO12
CO1 - Explain the influence of steady and variable stresses in machine component design.	3	2	1	1	3	2	2	2	2

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OBJECTIVES THE PHILOSOPHY SALIENT FEATURES

Objectives

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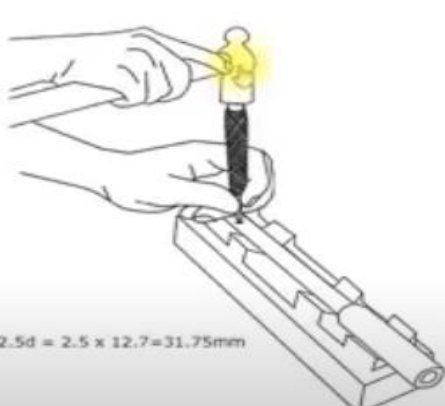
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TENSILE TEST ON MILD STEEL

STEP 3 Punch marks are made at a distance of $2.5d$.



Distance $2.5d = 2.5 \times 12.7 = 31.75\text{mm}$

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STEP II Observations and Calculations.

Result	Actual Value	Entered Value	Percentage Error
Yield Stress(N/mm^2)	347.52	356	-2.44
Tensile Strength(N/mm^2)	442.3	490	-10.78
Modulus of Elasticity(GPa)	209.65	212	-1.12
Percentage Elongation(%)	29.89	39	-30.48
Percentage reduction in area(%)	67.14	55	18.08

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